**PAYTON'S LiDAR SENSOR**
LASER SPEED · TF03 LiDAR · DESIGNED BY PAYTON PITTS

Vault approach report
TF03 LiDAR · 1000 Hz · **±0.03 s · ±0.1 m/s**

POLE VAULT · RUNWAY · SHORT APPROACH

# Demo Athlete

Nov 7, 2026 · 4:11 PM  Tripod 100 ft from the back of the box

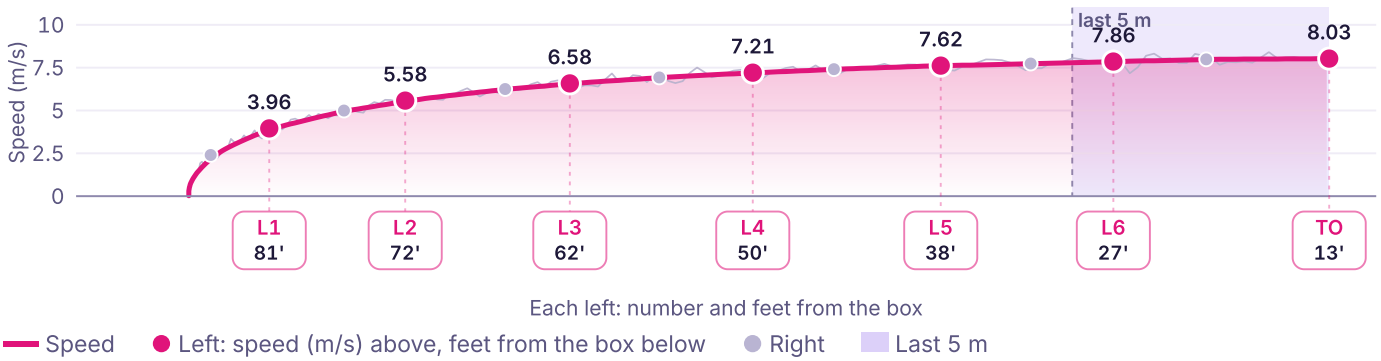
| Last 5 m into takeoff | Top speed | Lefts | Takeoff ≈ |
|---|---|---|---|
| **7.95 m/s** | **8.03 m/s** | **7** | **13' 5"** |
| 5 m before: 7.60 m/s | 91% of sprint top speed | 14 steps | out from the box |
| **+0.28 m/s** speeding up into the box<br>last ½ s before the plant | Start mark ≈ **86' 10"** from the box | Approach time **3.77 s** 72.8 ft run | Power **1024 W** 15.1 W/kg · peak horizontal |

## Summary · written by the app from this jump

- Demo's last 5 m into takeoff was **7.95 m/s**, built 0.36 into takeoff compared with the 5 m before it (7.60 m/s).
- Top speed on the runway was **8.03 m/s**, 91% of the best sprint top speed (8.83 m/s).
- Into the box: **+0.28 m/s**, speeding up into the box over the last half second before the plant.
- **7 lefts** (14 steps), counted back from the takeoff.

**NEXT SESSION** Still speeding up into the box: good. Keep the takeoff spot the same as speed rises, and check the mid mark stays put.

## Speed into the takeoff



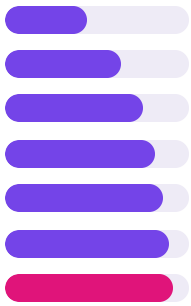
## Step rhythm

| # | FOOT | FROM BOX | STEP LENGTH | STEP TIME | SPEED (M/S) |
|---|---|---|---|---|---|
| 1 | R | 84' 10" | – | – | 2.40 |
| 2 | L1 | 81' 1" | 3' 9" | 0.35 s | 3.96 |
| 3 | R | 76' 3" | 4' 9" | 0.32 s | 5.00 |
| 4 | L2 | 72' 5" | 3' 11" | 0.22 s | 5.58 |
| 5 | R | 66' 0" | 6' 5" | 0.33 s | 6.25 |
| 6 | L3 | 61' 10" | 4' 2" | 0.19 s | 6.58 |
| 7 | R | 56' 2" | 5' 9" | 0.26 s | 6.93 |
| 8 | L4 | 50' 2" | 6' 0" | 0.26 s | 7.21 |
| 9 | R | 45' 0" | 5' 2" | 0.22 s | 7.41 |
| 10 | L5 | 38' 2" | 6' 10" | 0.28 s | 7.62 |
| 11 | R | 32' 5" | 5' 9" | 0.23 s | 7.74 |
| 12 | L6 | 27' 2" | 5' 3" | 0.21 s | 7.86 |
| 13 | R | 21' 3" | 5' 11" | 0.23 s | 7.99 |
| 14 | L7 · TO | 13' 5" | 7' 10" | 0.29 s | 8.03 |

## Speed you carry to the plant

| | |
|---|---|
| Sprint top speed | 8.83 m/s |
| Runway top speed | 8.03 m/s |
| Last 5 m | 7.95 m/s |
| Last 5 m as % of sprint top speed | 90% |
| Speed at the last 3 lefts | 7.62 → 7.86 → 8.03 |
| Step length: last 6 vs 6 before | 6' 2" vs 5' 2" |
| 33 → 16 ft out from the box | 7.94 m/s |

## Speed at each left




Bars are measured against sprint top speed. They should keep growing to the takeoff.

## How to use this as a vaulter

### Last 5 m into takeoff
The speed you bring to the plant, and the speed the pole has to work with. Faster here, with the same takeoff, means a stiffer pole and a higher grip. This is the number to chase.

### % of sprint top speed
How much of your max speed survives the pole carry. If it's low, the run-up is costing you speed: work on carry posture, relaxed arms and a steady rhythm. If it's high, the next jump in speed has to come from sprint training.

### Speed at each left
Should keep rising, or at least hold, all the way to takeoff. A dip at the last 2–3 lefts means slowing down to find the box or reaching for the plant.

### Step length (grey rights in between)
Steps should get a little shorter and quicker at the end, not longer. Overstriding in the last steps usually shows up as a speed drop into takeoff.

### Lefts, takeoff and start mark
The same lefts with a takeoff that keeps moving means the run-up is changing. Use the takeoff distance to move the start mark, and keep the left count the same while speed improves.

### Run-up length
The sprint report shows where you reach 98% of top speed. If your approach is shorter than that, you plant before you're at full speed. Add a left if you can still control it.